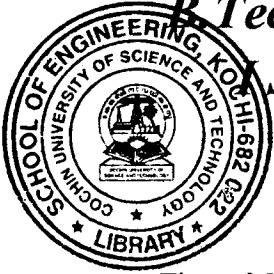


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**B.Tech. Degree II Semester Regular and Supplementary/
Semester Supplementary Examination May 2017**



CE/ME GE 15-1203 A (R) ENGINEERING GRAPHICS
(2015 Scheme)

Time : 3 Hours

Maximum Marks : 60

(5 × 12 = 60)

- I. $25 \times 10^{-3} \text{ m}^3$ of air at a pressure of 6 kg/cm^2 expands until its volume is $25 \times 10^{-2} \text{ m}^3$. The relation between pressure and volume is given by the formula $p v = 15 \times 10^{-2}$. Construct the expansion curve. Draw a tangent and normal to the curve at a point when the volume is $15 \times 10^{-2} \text{ m}^3$.

OR

- II. Draw one convolution of a logarithmic spiral given the shortest radius as 15 mm ratio of lengths of radius vectors enclosing an angle of 30° as 8:9. Draw a tangent and normal to the curve at any point and find the angle which the tangent makes with the radius vector.
- III. The plan mn of a straight line MN is 150 mm long and it makes an angle of 45° with XY . The end M is in the VP and 90 mm from HP . The end N is 30 mm from HP and the whole line lies in the fourth dihedral angle. Draw the projections of the line and determine its true length, traces and inclination.

OR

- IV. A regular hexagon $ABCDEF$ of side 30 mm has its plane inclined at 45° to the VP and its diagonal FC parallel to the VP and inclined at 45° to the HP . Draw its projections when the side DE is nearest to the VP and 10 mm in front of it.
- V. A right regular pentagonal pyramid side of base 45 mm and height 75 mm rests on one of its edges on the base in HP , the base being tilted up until the highest point in it is 60 mm above HP . Draw the projections of the pyramid when edge on which it rests is made parallel to the VP .

OR

- VI. A right circular cone height 105 mm diameter of base 65 mm lies on the horizontal plane with its axis plane parallel to the vertical plane. A section plane parallel to the VP cuts the cone at a distance of 20 mm from its axis. Draw the sectional elevation and true shape of the section.

- VII. A right circular cone of 70 mm diameter and 100 mm long rests in HP on its base. A point P initially situated at the extreme right end of the base moves around the surface of the cone and finally comes back to the starting point. Find the length of the shortest path, the point P will take in covering the distance along the surface of the cone. Also show the path in elevation and plan.

OR

- VIII. A hexagonal prism with its axis vertical, edges of the base 40 mm and height 135 mm pierces a square prism side of base 40 mm and height 135 mm. The axis of the square prism is inclined at 30° to the HP and parallel to the VP. The elevation of their axes bisect each other and the distance between their axes seen in plan is 15 mm. Project an elevation and plan showing the curves of penetration in each view.

- IX. A hollow cube of 30 mm and 40 mm of inside and outside edges is mounted centrally over a cylindrical block of 70 mm diameter and 15 mm thick. Draw the isometric projection of the solids.

OR

- X. A regular pentagonal pyramid edge of base 1000 mm and height 1000 mm stands on its base in HP such that one of its sides touches the picture plane. Take the height of the eye 1500 mm and its distance 2000 mm, project the perspective projection of the pyramid.